

Coor # 8 : Cinématique de rotation

* Rappel
MCU/MCUA

* Systeme contraintes
Poulie/engrenage

Relations paramètres
Linéaire / angulaires

* Applications
Position angulaire
↓
 θ, ω, α
↓
vitesse angulaire
↓
 $\theta = \frac{L}{R}$

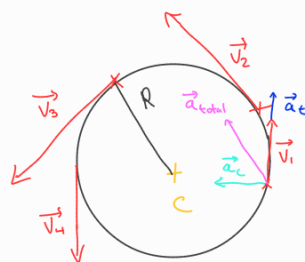
$$\theta = \frac{L}{R}$$

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t}$$

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega}{\Delta t}$$

INTRA 2

$$\begin{aligned} L &= \theta \cdot R \\ v &= \omega \cdot R \\ a_t &= \alpha \cdot R \end{aligned}$$



$$\vec{a}_{total} = \vec{a}_c + \vec{a}_t$$

Grandeur
 $a_{total} = \sqrt{a_c^2 + a_t^2}$

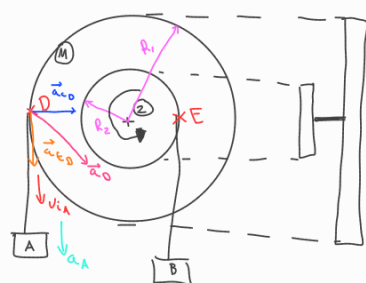
a_t = Variation de grandeur de $\vec{v}$

a_c = Variation de direction de $\vec{v}$

$$a_t = \frac{\Delta v}{\Delta t}$$

$$a_c = \frac{v^2}{R}$$

application #1 Moodle



Données

$$R_1 = 200 \text{ mm}$$

$$R_2 = 90 \text{ mm}$$

$$a_A = C^{te} = 75 \text{ mm/s}^2$$

$$v_{iA} = 20 \text{ mm/s}$$

1) on sait $a_t = \alpha_D R_1 = 75 \text{ mm/s}^2 = C^{te}$

$$\Rightarrow \alpha_D = 0,375 \text{ rad/s}^2 = C^{te} \Rightarrow \text{MCUA}$$

Initial	Final
$\theta_{i_{R_1}} = 0 \text{ tour}$	$\theta_{f_{R_1}} = ?$
$\omega_{i_{R_1}} = 0,6 \text{ rad/s}$	$\omega_{f_{R_1}} = ?$
$t_{i_{R_1}} = 0 \text{ s}$	$t_{f_{R_1}} = 6 \text{ s}$

* on sait que

$$v_{iA} = \omega_{i_{R_1}} \cdot R$$

$$\Rightarrow \omega_{i_{R_1}} = 0,6 \text{ rad/s}$$

$$\textcircled{1} \theta_2 = \theta_1 + \omega_1 (t_2 - t_1) + \frac{1}{2} \alpha (t_2 - t_1)^2$$

$$\textcircled{2} \omega_2 = \omega_1 + \alpha (t_2 - t_1)$$

$$\textcircled{1} \theta_{f_{R_1}} = 0 + 0,6 (6 - 0) + \frac{1}{2} \cdot 0,375 (6 - 0)^2$$

$$\textcircled{2} \omega_{f_{R_1}} = 0,6 + 0,375 (6 - 0)$$

$$\theta_{f_{R_1}} = 10,35 \text{ rad}$$

$$\omega_{f_{R_1}} = 2,85 \text{ rad/s}$$

$$2\pi \text{ rad} = 1 \text{ tour}$$

$$10,35 \text{ rad} = \frac{10,35}{2\pi} \text{ tour}$$

$$\theta_{f_{R_1}} = 1,647 \text{ tour}$$

2) Roues concentriques

$$\Rightarrow \omega_{R_1} = \omega_{R_2}$$

$$r_{R_1} = r_{R_2}$$

$$\theta_{R_1} = \theta_{R_2}$$

$$\theta_{t_{R_2}} = \theta_{t_{R_1}}$$

$$\omega_{t_{R_1}} = \omega_{t_{R_2}}$$

De plus la vitesse B

$$V_B = V_E$$

$$V_B = \omega_{t_{R_2}} \cdot R_2 = 256,5 \text{ mm/s}$$

$$L = \theta \cdot R$$

$$L_B = \theta_{t_{R_2}} \cdot R_2 = 10,35 \cdot 90 \text{ mm} = 931,5 \text{ mm}$$

3) $\vec{a} = \vec{a}_c + \vec{a}_t$ @ $t = 0 \text{ sec}$

$$\vec{a}_0 = \vec{a}_{c_0} + \vec{a}_{t_0}$$

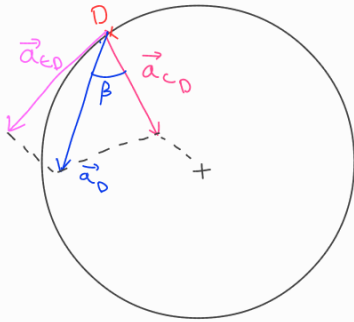
$$@ t = 0 \text{ sec} \quad a_{t_0} = a_A = 75 \text{ mm/s}^2$$

$$\vec{a}_{c_0} = \frac{V_0^2}{R_1} @ t = 0 \text{ s} \quad a \quad V_0 = V_{i_A} = 120 \text{ mm/s}$$

$$\vec{a}_{c_0} = 72 \text{ mm/s}^2$$

Grandeur : $a_0 = \sqrt{a_{c_0}^2 + a_{t_0}^2} = 103,97 \text{ mm/s}^2$

Direction : $\tan \beta = \frac{a_{t_0}}{a_{c_0}} \Rightarrow \beta = 46,17^\circ$



Résumé

$$L = \Theta \cdot R$$

$$V = \omega \cdot R$$

$$a_t = \alpha \cdot R$$

$$\vec{a}_{\text{total}} = \vec{a}_c + \vec{a}_t$$

$$a_c = \frac{V^2}{R} \quad a_t = \frac{\Delta V}{\Delta t}$$

$$\Theta_1 \cdot R_1 = \Theta_2 \cdot R_2$$

$$\omega_1 \cdot R_1 = \omega_2 \cdot R_2$$

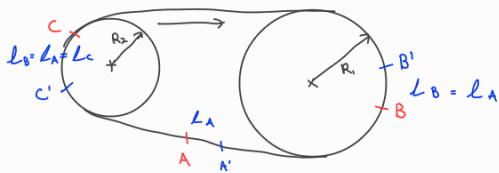
$$\alpha_1 \cdot R_1 = \alpha_2 \cdot R_2$$

dans le cas

courroie / engrenage

2. Systèmes Contraints : Courroie et engrenage /

Preons situation suivante :



2.1 Description du mouvement d'une poulie (R2) connaissant le mouvement de l'autre poulie (R1)

Hyp : Courroie non-élastique et sans glissement

$$L_A = L_B = L_C$$

$$AA' = BB' = CC' \Rightarrow L_B = L_C$$

On sait $L = \Theta \cdot R$ $\boxed{\Theta_1 \cdot R_1 = \Theta_2 \cdot R_2}$

RMQ : si $R_1 > R_2$ alors $\Theta_1 < \Theta_2$

Vitesse : on sait

$$V_A = V_B = V_C$$

$$V_B = \omega_1 \cdot R_1 \Rightarrow \boxed{\omega_1 \cdot R_1 = \omega_2 \cdot R_2}$$

$$V_C = \omega_2 \cdot R_2$$

RMQ : si $R_1 > R_2$ alors $\omega_1 > \omega_2$

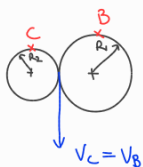
accélération : $a_A = a_{tB} = a_{tC}$

On sait $\begin{cases} a_{tB} = \alpha_1 \cdot R_1 \\ a_{tC} = \alpha_2 \cdot R_2 \end{cases}$

$$\Rightarrow \boxed{\alpha_1 \cdot R_1 = \alpha_2 \cdot R_2}$$

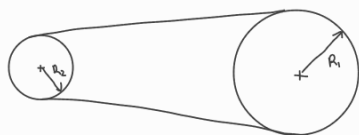
RMQ : si $R_1 > R_2$ alors $\alpha_1 > \alpha_2$

engrenage :



Si $R_1 > R_2$

Exercice #2 moodle



Données

$$R_1 = 120 \text{ mm}$$

$$R_2 = 45 \text{ mm}$$

$$\omega_1 = 12 \text{ tour/s}$$

$$\alpha_1 = 8 \text{ rad/s}^2$$

a) $\omega_2 \cdot R_2 = \omega_1 \cdot R_1$

$$\omega \text{ rad/s} \quad v = \omega \cdot R$$

$$\omega [\text{rad/s}]$$

$$v [\text{m/s}] \quad R : [\text{m}]$$

$$\omega_2 = \frac{12 \times 2\pi \cdot R_1}{R_2} = 201,06 \text{ rad/s}$$

b) $\alpha_2 \cdot R_2 = \alpha_1 \cdot R_1$

$$\alpha_2 = \frac{8 \times 120}{45} = 21,33 \text{ rad/s}^2$$

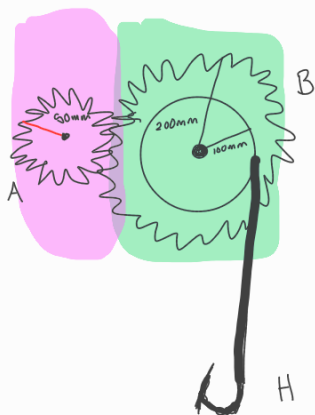
c) $\vec{a}_2 = \vec{a}_{c2} + \vec{a}_{t2} @ \omega_1 = 12 \text{ tour/s}$

$$a_{c2} = \frac{v_2^2}{R_2} \quad \text{avec } v_2 = \omega_2 \cdot R_2 = 9047,79 \text{ mm/s}$$

$$a_{c2} = \frac{(9047,79)^2}{45} = 1,819 \cdot 10^6 \text{ mm/s}^2$$

$$a_{t2} = \alpha_2 \cdot R_2 = 960 \text{ mm/s}^2$$

Exercice #3 moodle



■ = engrenage B et poulie = concentrique

■ = engrenage B et A = courroie et poulies

a) $a_{tA} = \alpha_A \cdot R_A = 2 \cdot 50 = 100 \text{ mm/s}^2 \rightarrow C^{te}$

b) on sait que $a_{tA} = a_{tB}$

$$\alpha_B \cdot R_B = \alpha_A \cdot R_A$$

$$\alpha_B = \frac{\alpha_A \cdot R_A}{R_B} = 0,5 \text{ rad/s}^2 \rightarrow C^{te}$$

c) B et Poulie sont concentrique Poulie = P

$$\alpha_B = \alpha_P \quad \text{et} \quad a_{tP} = \alpha_P \cdot R_P$$

$$a_{tP} = 0,5 \cdot 100 = 50 \text{ mm/s}^2 \rightarrow C^{te}$$

d) comme a_{tP} est constante le crochet possède un MRUA dont l'accélération est a_{tP}

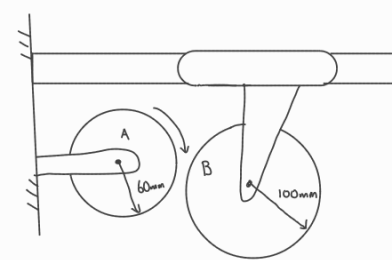
$$① y_2 = y_1 + v_1(t_2 - t_1) + \frac{1}{2} a(t_2 - t_1)^2$$

①	$a = a_{tP}$	②
$y_1 = 0$		$y_2 = ?$
$v_1 = 0$		$v_2 = ?$
$t_1 = 0$		$t_2 = 10 \text{ s}$

$$y_2 = 0 + 0(10-0) + \frac{1}{2} \cdot 50(10-0)^2$$

$$y_2 = 2500 \text{ mm}$$

Exercice #4 moodle



Données

$$R_A = 60\text{mm}$$

$$R_B = 100\text{mm}$$

$$\omega_{1B} = 0$$

$$\omega_{1A} = 450 \text{ tour/min}$$

$$t_1 = 0$$

$$@ t_L = 6\text{s} \quad \omega_{2A} = 140 \text{ tour/min}$$

a) α_A ? en rad/s^2

①	α_A	②
$\theta_{1A} = 0$		$\theta_{2A} = ?$
$\omega_{1A} = 450 \text{ tour/min} \cdot \frac{2\pi}{60}$		$\omega_{2A} = 140 \text{ tour/min} \times \frac{2\pi}{60}$
$\alpha_{1A} = ?$		$\alpha_{2A} = \alpha_{1A}$
$t_{1A} = 0$		$t_{2A} = 6\text{s}$

$$\textcircled{1} \theta_2 = \theta_1 + \omega_{1A}(t_2 - t_1) + \frac{1}{2} \cdot \alpha_A (t_2 - t_1)^2$$

$$\textcircled{2} \omega_{2A} = \omega_{1A} + \alpha_A (t_2 - t_1)$$

$$\alpha_A = -5.41 \text{ rad/s}^2$$

$$\theta_{2A} = 185.35 \text{ rad}$$

b) $\alpha_B = ?$

①	②
$\theta_{1B} = 0$	$\theta_{2B} = ?$
$\omega_{1B} = 0$	$\omega_{2B} = ?$
$t_1 = 0$	$t_2 = 6\text{s}$
$\alpha_{1B} = ?$	$\alpha_{2B} = \alpha_{1B}$

on sait que à $t = 6\text{s}$ plus de glissement

$$\Rightarrow v_{2B} = v_{2A} \Rightarrow \omega_{2A} \cdot R_A = \omega_{2B} \cdot R_B$$

$$\hookrightarrow \omega_{2B} = 8.796 \text{ rad/s}$$